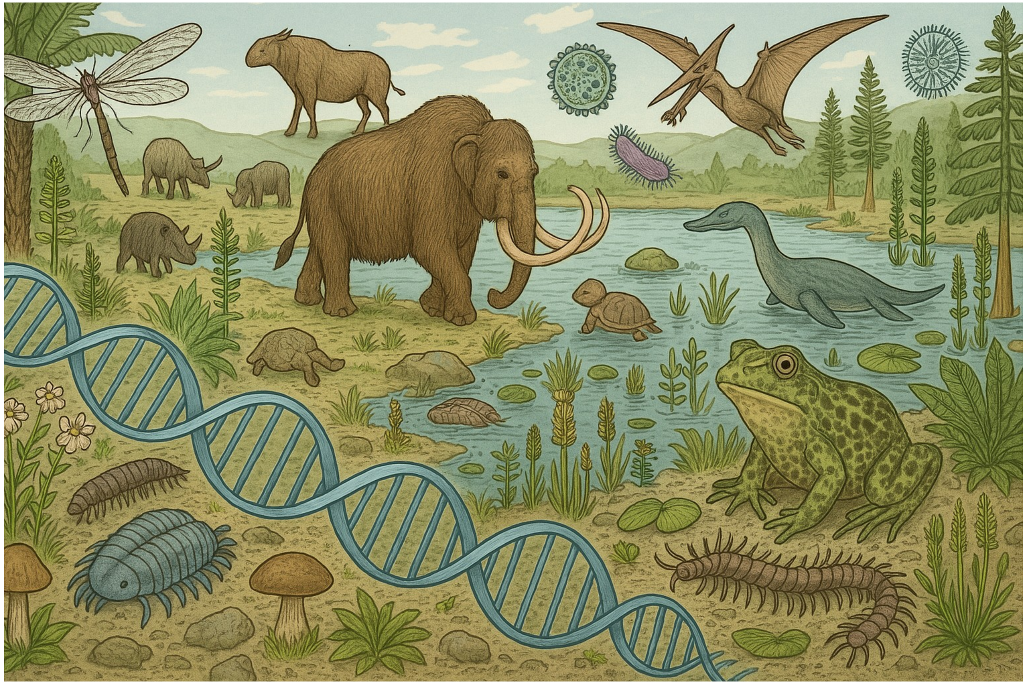
ANCIENT METAGENOMICS

Physalia course, online, 2-4 June 2026

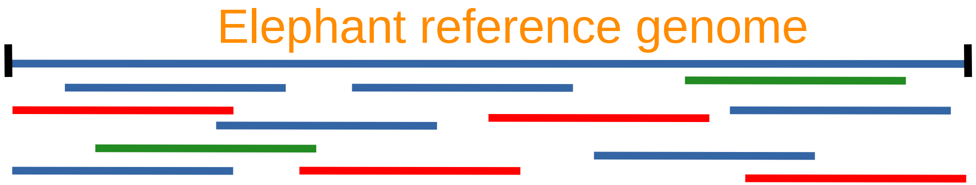
Decontamination of metagenomic data and eukaryotic references

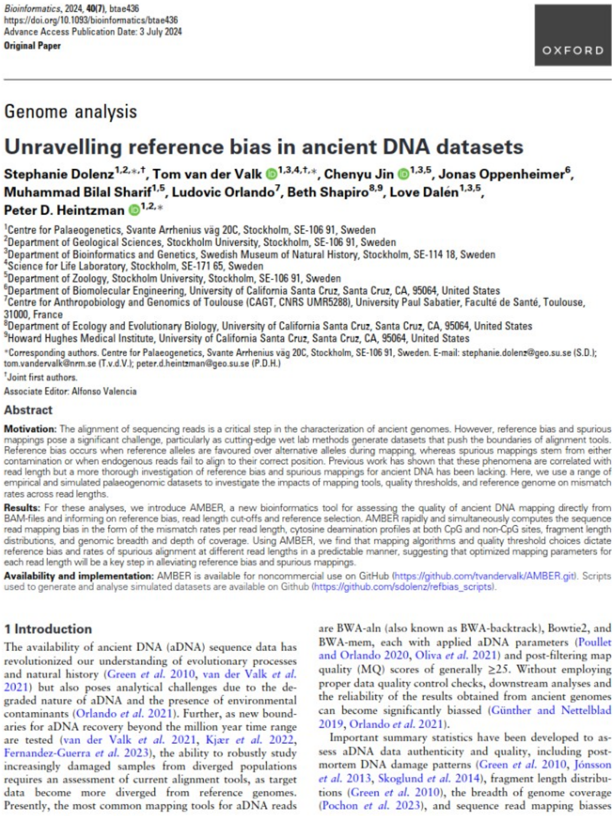
Nikolay Oskolkov, Lund University, NBIS SciLifeLab



Physalia Courses

Remove short (likely contaminant) reads





Bioinformatics, 2024, **40(7)**, btae436
https://doi.org/10.1093/bioinformatics/btae436
Advance Access Publication Date: 3 July 2024
Original Paper

OXFORD

Genome analysis

Unravelling reference bias in ancient DNA datasets

Stephanie Dolenz[1,2,*,†], Tom van der Valk[1,3,4,†,*], Chenyu Jin[1,3,5], Jonas Oppenheimer[6], Muhammad Bilal Sharif[1,5], Ludovic Orlando[7], Beth Shapiro[8,9], Love Dalén[1,3,5], Peter D. Heintzman[1,2,*]

[1]Centre for Palaeogenetics, Svante Arrhenius väg 20C, Stockholm, SE-106 91, Sweden
[2]Department of Geological Sciences, Stockholm University, Stockholm, SE-106 91, Sweden
[3]Department of Bioinformatics and Genetics, Swedish Museum of Natural History, Stockholm, SE-114 18, Sweden
[4]Science for Life Laboratory, Stockholm, SE-171 65, Sweden
[5]Department of Zoology, Stockholm University, Stockholm, SE-106 91, Sweden
[6]Department of Biomolecular Engineering, University of California Santa Cruz, Santa Cruz, CA, 95064, United States
[7]Centre for Anthropobiology and Genomics of Toulouse (CAGT, CNRS UMR5288), University Paul Sabatier, Faculté de Santé, Toulouse, 31000, France
[8]Department of Ecology and Evolutionary Biology, University of California Santa Cruz, Santa Cruz, CA, 95064, United States
[9]Howard Hughes Medical Institute, University of California Santa Cruz, Santa Cruz, CA, 95064, United States

*Corresponding authors. Centre for Palaeogenetics, Svante Arrhenius väg 20C, Stockholm, SE-106 91, Sweden. E-mail: stephanie.dolenz@geo.su.se (S.D.); tom.vandervalk@nrm.se (T.v.d.V.); peter.d.heintzman@geo.su.se (P.D.H.)

†Joint first authors.

Associate Editor: Alfonso Valencia

Abstract

Motivation: The alignment of sequencing reads is a critical step in the characterization of ancient genomes. However, reference bias and spurious mappings pose a significant challenge, particularly as cutting-edge wet lab methods generate datasets that push the boundaries of alignment tools. Reference bias occurs when reference alleles are favoured over alternative alleles during mapping, whereas spurious mappings stem from either contamination or when endogenous reads fail to align to their correct position. Previous work has shown that these phenomena are correlated with read length but a more thorough investigation of reference bias and spurious mappings for ancient DNA has been lacking. Here, we use a range of empirical and simulated palaeogenomic datasets to investigate the impacts of mapping tools, quality thresholds, and reference genome on mismatch rates across read lengths.

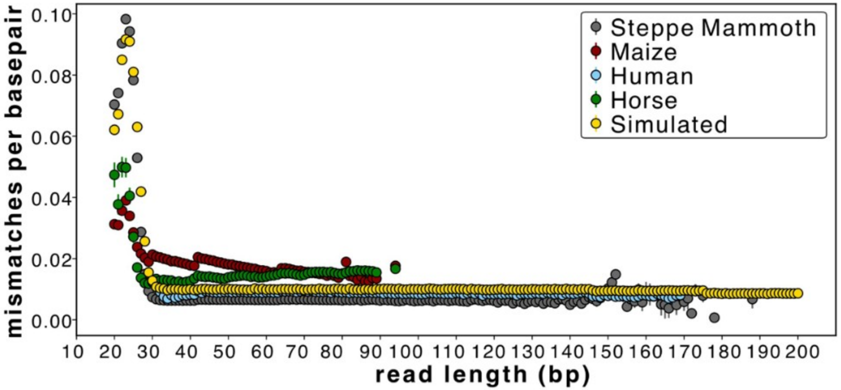
Results: For these analyses, we introduce AMBER, a new bioinformatics tool for assessing the quality of ancient DNA mapping directly from BAM-files and informing on reference bias, read length cut-offs and reference selection. AMBER rapidly and simultaneously computes the sequence read mapping bias in the form of the mismatch rates per read length, cytosine deamination profiles at both CpG and non-CpG sites, fragment length distributions, and genomic breadth and depth of coverage. Using AMBER, we find that mapping algorithms and quality threshold choices dictate reference bias and rates of spurious alignment at different read lengths in a predictable manner, suggesting that optimized mapping parameters for each read length will be a key step in alleviating reference bias and spurious mappings.

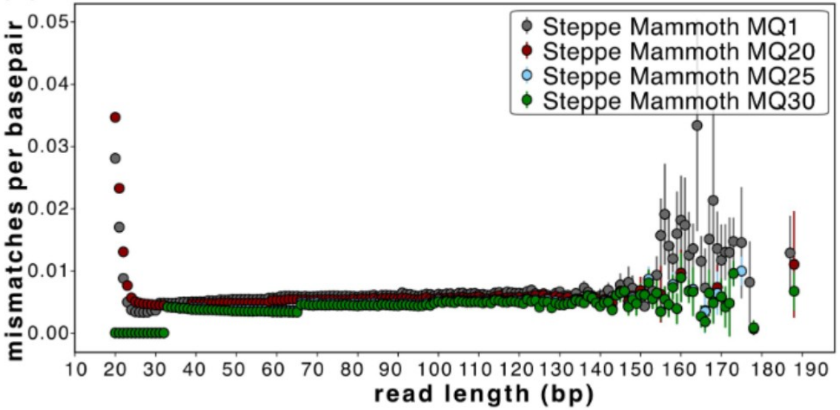
Availability and implementation: AMBER is available for noncommercial use on GitHub (https://github.com/tvandervalk/AMBER.git). Scripts used to generate and analyse simulated datasets are available on Github (https://github.com/sdolenz/refbias_scripts).

1 Introduction

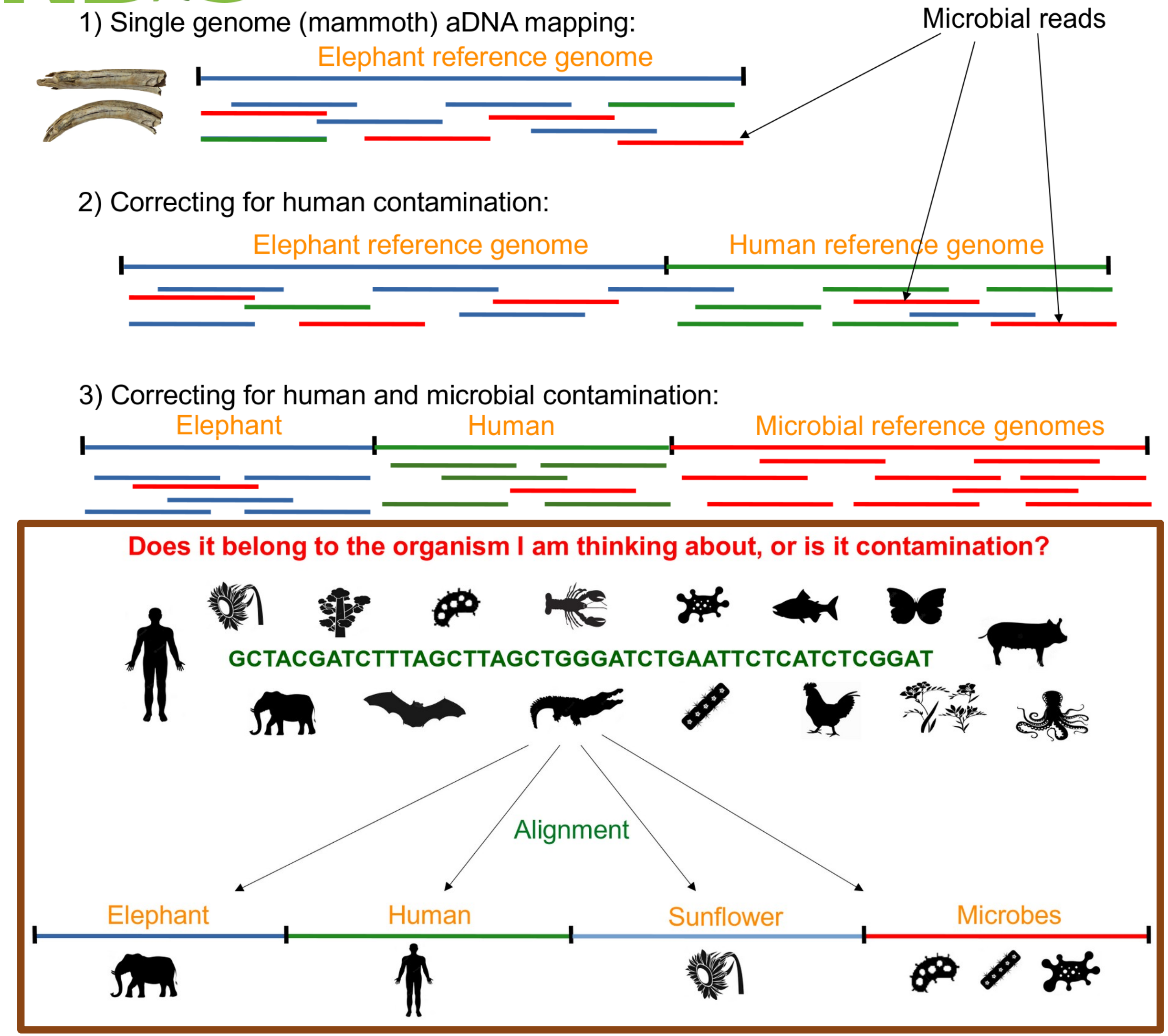
The availability of ancient DNA (aDNA) sequence data has revolutionized our understanding of evolutionary processes and natural history (Green *et al.* 2010, van der Valk *et al.* 2021) but also poses analytical challenges due to the degraded nature of aDNA and the presence of environmental contaminants (Orlando *et al.* 2021). Further, as new boundaries for aDNA recovery beyond the million year time range are tested (van der Valk *et al.* 2021, Kjær *et al.* 2022, Fernandez-Guerra *et al.* 2023), the ability to robustly study increasingly damaged samples from diverged populations requires an assessment of current alignment tools, as target data become more diverged from reference genomes. Presently, the most common mapping tools for aDNA reads are BWA-aln (also known as BWA-backtrack), Bowtie2, and BWA-mem, each with applied aDNA parameters (Poullet and Orlando 2020, Oliva *et al.* 2021) and post-filtering map quality (MQ) scores of generally ≥25. Without employing proper data quality control checks, downstream analyses and the reliability of the results obtained from ancient genomes can become significantly biassed (Günther and Nettelblad 2019, Orlando *et al.* 2021).

Important summary statistics have been developed to assess aDNA data authenticity and quality, including post-mortem DNA damage patterns (Green *et al.* 2010, Jónsson *et al.* 2013, Skoglund *et al.* 2014), fragment length distributions (Green *et al.* 2010), the breadth of genome coverage (Pochon *et al.* 2023), and sequence read mapping biasses





Competitive mapping or ultra-competitive?



Feuerborn *et al. BMC Genomics* (2020) 21:844
https://doi.org/10.1186/s12864-020-07229-y

BMC Genomics

METHODOLOGY ARTICLE Open Access

Competitive mapping allows for the identification and exclusion of human DNA contamination in ancient faunal genomic datasets

Tatiana R. Feuerborn[1,2,3,4*], Eleftheria Palkopoulou[3], Tom van der Valk[3,4], Johanna von Seth[3,4,5], Arielle R. Munters[6], Patrícia Pečnerová[7], Marianne Dehasque[3,4,5], Irene Ureña[8], Erik Ersmark[3,4], Vendela Kempe Lagerholm[2,4], Maja Krzewińska[2,4], Ricardo Rodríguez-Varela[2,4], Anders Götherström[2,4], Love Dalén[3,4,5] and David Díez-del-Molino[3,4,5*]

Abstract

Background: After over a decade of developments in field collection, laboratory methods and advances in high-throughput sequencing, contamination remains a key issue in ancient DNA research. Currently, human and microbial contaminant DNA still impose challenges on cost-effective sequencing and accurate interpretation of ancient DNA data.

Results: Here we investigate whether human contaminating DNA can be found in ancient faunal sequencing datasets. We identify variable levels of human contamination, which persists even after the sequence reads have been mapped to the faunal reference genomes. This contamination has the potential to affect a range of downstream analyses.

Conclusions: We propose a fast and simple method, based on competitive mapping, which allows identifying and removing human contamination from ancient faunal DNA datasets with limited losses of true ancient data. This method could represent an important tool for the ancient DNA field.

Keywords: Ancient DNA, DNA contamination removal, Palaeogenomics, Competitive mapping

Background

Right after the death of an organism, microbial communities colonize the decomposing tissues and together with enzymes from the organism they start degrading the DNA molecules [1–3]. DNA degradation is dependent on time and environmental variables such as temperature but also humidity and acidity [4]. Even though the specific model for DNA decay is still debated and it is likely multifactorial [4], the consequence is that ancient remains typically contain very few molecules of endogenous DNA and these sequences are characterized by short fragment sizes [5].

A second major challenge of ancient DNA research is contamination from exogenous sources [6, 7]. Environmental DNA molecules in the soil matrix the ancient sample was recovered from can easily overwhelm the small amounts of endogenous DNA. This is also true for DNA from people who collected and handled the samples in the field and/or museum collections [8, 9]. While the use of Polymerase Chain Reaction (PCR) technology allowed ancient DNA research to overcome low concentration problems, the sensitivity of the PCR has made it

* Correspondence: tatianafeuerborn@palaeome.org; diez.molino@gmail.com
[1]Globe Institute, University of Copenhagen, Copenhagen, Denmark
[3]Department of Bioinformatics and Genetics, Swedish Museum of Natural History, Stockholm, Sweden
Full list of author information is available at the end of the article

Decontamination with Kraken prior to mapping

OXFORD

Briefings in Bioinformatics, 2025, 26(1), bbae646
https://doi.org/10.1093/bib/bbae646
Problem Solving Protocol

Filtering out the noise: metagenomic classifiers optimize ancient DNA mapping

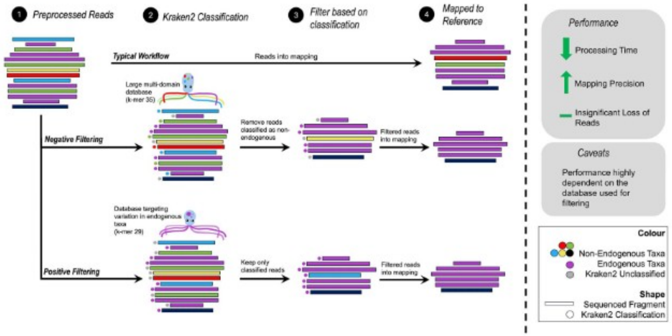
Shyamsundar Ravishankar[1,*], Vilma Perez[1,2], Roberta Davidson[1], Xavier Roca-Rada[1,3], Divon Lan[1,4], Yassine Souilmi[1,5,6,‡], Bastien Llamas[1,2,5,6,‡]

[1]Australian Centre for Ancient DNA (ACAD) and The Environment Institute, The School of Biological Sciences, University of Adelaide, Adelaide, SA, Australia
[2]Centre of Excellence for Australian Biodiversity and Heritage, University of Adelaide, Adelaide, SA, Australia
[3]Faculty of Arts and Humanities, University of Coimbra, Coimbra, Portugal
[4]Genozip Limited, Hong Kong
[5]National Centre for Indigenous Genomics, Australian National University, Canberra, ACT, Australia
[6]Indigenous Genomics, Telethon Kids Institute, Adelaide, SA, Australia
*Corresponding author. Australian Centre for Ancient DNA (ACAD) and The Environment Institute, The School of Biological Sciences, The University of Adelaide, North Terrace, Adelaide, South Australia, 5008, Australia. E-mail: shyamsundar.ravishankar@adelaide.edu.au
‡Yassine Souilmi and Bastien Llamas are equally contributing last authors.

Abstract

Contamination with exogenous DNA presents a significant challenge in ancient DNA (aDNA) studies of single organisms. Failure to address contamination from microbes, reagents, and present-day sources can impact the interpretation of results. Although field and laboratory protocols exist to limit contamination, there is still a need to accurately distinguish between endogenous and exogenous data computationally. Here, we propose a workflow to reduce exogenous contamination based on a metagenomic classifier. Unlike previous methods that relied exclusively on DNA sequencing reads mapping specificity to a single reference genome to remove contaminating reads, our approach uses *Kraken2*-based filtering before mapping to the reference genome. Using both simulated and empirical shotgun aDNA data, we show that this workflow presents a simple and efficient method that can be used in a wide range of computational environments—including personal machines. We propose strategies to build specific databases used to profile sequencing data that take into consideration available computational resources and prior knowledge about the target taxa and likely contaminants. Our workflow significantly reduces the overall computational resources required during the mapping process and reduces the total runtime by up to ~94%. The most significant impacts are observed in low endogenous samples. Importantly, contaminants that would map to the reference are filtered out using our strategy, reducing false positive alignments. We also show that our method results in a negligible loss of endogenous data with no measurable impact on downstream population genetics analyses.

Graphical Abstract



Keywords: ancient DNA; contamination; filtering, metagenomic classifiers; Kraken2

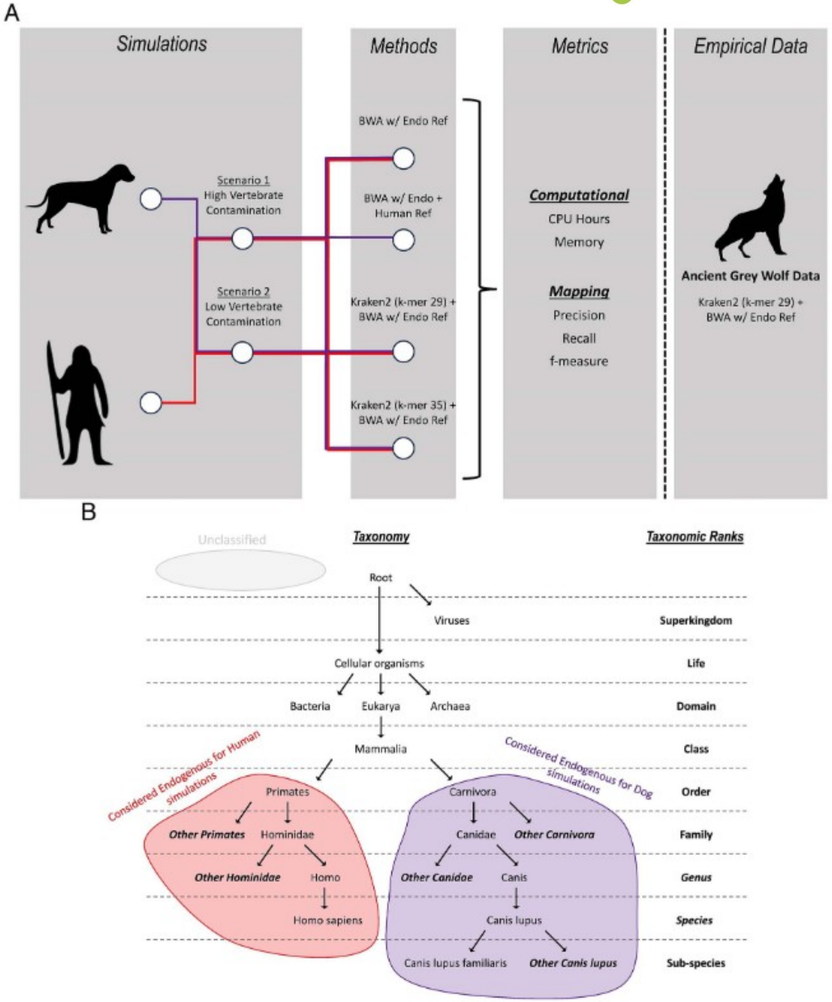


Figure 1. (A) Workflow of types of simulated data, different methods applied to the simulated data, and metrics collected. The best-performing method is applied to empirical data. (B) Ancient human reads classified at the order Primates or lower taxonomic ranks are considered endogenous reads and hence retained (red), similarly for ancient dog reads at the order Carnivora or lower (purple). Unclassified reads represented as grey are reads that could not be assigned a taxonomy.

Outline

- Contamination problem
  - DNA of modern (microbial) organisms in ancient / historical samples
  - Animal / plant / bird / fish etc. references include microbial contamination

- Microbiome contamination correction
  - Decontamination via negative controls (blanks)
  - Similarity to expected microbiome source (microbial source tracking)

Contamination problem

Modern microbial and animal DNA contamination

BMC Part of Springer Nature

BMC Biology

Home About Articles Submission Guidelines

Research article | Open Access | Published: 12 November 2014

Reagent and laboratory contamination can critically impact sequence-based microbiome analyses

Susannah J Salter, Michael J Cox, Elena M Turek, Szymon T Calus, William O Cookson, Miriam F Moffatt, Paul Turner, Julian Parkhill, Nicholas J Loman & Alan W Walker

BMC Biology 12, Article number: 87 (2014) | Cite this article

Abstract

Background

The study of microbial communities has been revolutionised in recent years by the widespread adoption of culture independent analytical techniques such as 16S rRNA gene sequencing and metagenomics. One potential confounder of these sequence-based approaches is the presence of contamination in DNA extraction kits and other laboratory reagents.

Results

In this study we demonstrate that contaminating DNA is ubiquitous in commonly used DNA extraction kits and other laboratory reagents, varies greatly in composition between different kits and kit batches, and that this contamination critically impacts results obtained from samples containing a low microbial biomass. Contamination impacts both PCR-based 16S rRNA gene surveys and shotgun metagenomics. We provide an extensive list of potential

ELSEVIER

Journal of Archaeological Science

Volume 34, Issue 9, September 2007, Pages 1361-1366

Animal DNA in PCR reagents plagues ancient DNA research

Jennifer A. Leonard [a, b, c], Orin Shanks [d, 1], Michael Hofreiter [e], Eva Kreuz [e], Larry Hodges [d], Walt Ream [d], Robert K. Wayne [b], Robert C. Fleischer [a]

Show more

+ Add to Mendeley Share Cite

https://doi.org/10.1016/j.jas.2006.10.023 Get rights and content

Abstract

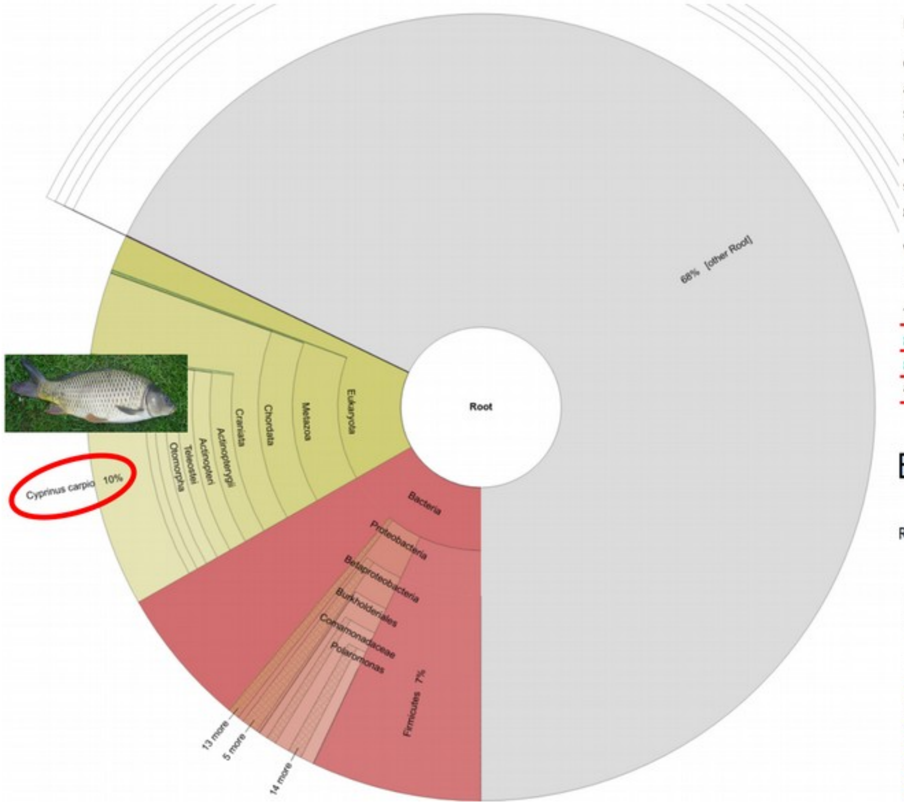
Molecular archaeology brings the tools of molecular biology to bear on fundamental questions in archaeology, anthropology, evolution, and ecology. Ancient DNA research is becoming widespread as evolutionary biologists and archaeologists discover the power of the polymerase chain reaction (PCR) to amplify DNA from ancient plant and animal remains. However, the extraordinary susceptibility of PCR to contamination by extraneous DNA is not widely appreciated. We report the independent observation of DNA from domestic animals in PCR reagents and ancient samples in four separate laboratories. Since PCR conditions used in ancient DNA analyses are extremely sensitive, very low concentrations of contaminating DNA can cause false positives. Previously unidentified animal DNA in reagents can confound ancient DNA research on certain domestic animals, especially cows, pigs, and chickens.

Previous article in issue Next article in issue

Keywords

Sus scrofa; *Bos taurus*; *Gallus gallus*; Deoxynucleoside triphosphates

Contaminated reference databases



It turns out the Carp genome is full of Illumina adapters.

One of the first things we teach people in our NGS courses is how to remove adapters. It's not hard – we use CutAdapt, but many other tools exist. It's simple, but really important – with De Bruijn graphs you will get paths through the graphs converging on kmers from adapters; and with OLC assemblers you will get spurious overlaps. With gap-fillers, it's possible to fill the gaps with sequences ending in adapters, and this may be what happened in the Carp genome.

Why then are we finding such elementary mistakes in such important papers? Why aren't reviewers picking up on this? It's frustrating.

This is a separate, but related issue, to genomic contamination – the Wheat genome has PhiX in it; tons of bacterial genomes do too; and lots of bacterial genes were problematically included in the Tardigrade genome and declared as horizontal gene transfer.

Bioinformatics Bits and Bobs

Rare and random blog posts about bioinformatics, genomics and evolution.

Monday, 29 September 2014

Why you should QC your reads AND your assembly

The genome sequence of the Common Carp *Cyprinus carpio* was published in Nature last week. By coincidence, I was doing some QC on some domesticated Ferret (*Mustela ptorius furo*) reads, which had thrown some kmer warnings in the FastQC tool. I blasted the kmers in NCBI and was quite perplexed by the number of hits that I found in the carp genome. Nearly all of the first 150 hits were all from the carp genome. Anyway, I looked a bit further into my odd kmers and it turns out that they were the ends of some Illumina adapter sequences that had presumably been incorporated into the paired-reads on the shorter ends of the insert size. This then took me back to the Carp Genome - what had creeped into that?

Microbial contamination of RefSeq genomes

Tibetan antelope (*Pantholops hodgsonii*) turns up in every metagenomic dataset 🤔

Pantholops hodgsonii as well as *Bos mutus / taurus* and a few other mammals have severe microbial contamination (Oskolkov et al. 2025)

Eukaryotic contamination in microbial references

RefSeq genome of the common soil bacterium *Achromobacter denitrificans* contains the entire chicken ovalbumin gene!

Spirometra erinaceieuropaei (tapeworm) seems to have human contamination (Jensen et al., Nat. Comm. 2019)

Microbiome Contamination Correction

Contaminants found on negative controls

Table 1 List of contaminant genera detected in sequenced negative `blank' controls

From: Reagent and laboratory contamination can critically impact sequence-based microbiome analyses

Phylum	List of constituent contaminant genera
Proteobacteria	Alpha-proteobacteria:
	Afipia, *Aquabacterium*[e], *Asticcacaulis*, *Aurantimonas*, *Beijerinckia*, *Bosea*, *Bradyrhizobium*[d], *Brevundimonas*[c], *Caulobacter*, *Craurococcus*, *Devosia*, *Hoefleda*[e], *Mesorhizobium*, *Methylobacterium*[c], *Novosphingobium*, *Ochrobactrum*, *Paracoccus*, *Pedomicrobium*, *Phyllobacterium*[e], *Rhizobium*[c,d], *Roseomonas*, *Sphingobium*, *Sphingomonas*[c,d,e], *Sphingopyxis*
	Beta-proteobacteria:
	Acidovorax[c,e], *Azoarcus*[e], *Azospira*, *Burkholderia*[d], *Comamonas*[c], *Cupriavidus*[c], *Curvibacter*, *Delftia*[e], *Duganella*[a], *Herbaspirillum*[a,c], *Janthinobacterium*[e], *Kingella*, *Leptothrix*[a], *Limnobacter*[e], *Massilia*[c], *Methylophilus*, *Methyloversatilis*[e], *Oxalobacter*, *Pelomonas*, *Polaromonas*[e], *Ralstonia*[b,c,d,e], *Schlegelella*, *Sulfuritalea*, *Undibacterium*[e], *Variovorax*
	Gamma-proteobacteria:
	Acinetobacter[a,d,c], *Enhydrobacter*, *Enterobacter*, *Escherichia*[a,c,d,e], *Nevskia*[e], *Pseudomonas*[b,d,e], *Pseudoxanthomonas*, *Psychrobacter*, *Stenotrophomonas*[a,b,c,d,e], *Xanthomonas*[b]
Actinobacteria	*Aeromicrobium*, *Arthrobacter*, *Beutenbergia*, *Brevibacterium*, *Corynebacterium*, *Curtobacterium*, *Dietzia*, *Geodermatophilus*, *Janibacter*, *Kocuria*, *Microbacterium*, *Micrococcus*, *Microlunatus*, *Patulibacter*, *Propionibacterium*[e], *Rhodococcus*, *Tsukamurella*
Firmicutes	*Abiotrophia*, *Bacillus*[b], *Brevibacillus*, *Brochothrix*, *Facklamia*, *Paenibacillus*, *Streptococcus*
Bacteroidetes	*Chryseobacterium*, *Dyadobacter*, *Flavobacterium*[d], *Hydrotalea*, *Niastella*, *Olivibacter*, *Pedobacter*, *Wautersiella*
Deinococcus-Thermus	*Deinococcus*
Acidobacteria	Predominantly unclassified Acidobacteria Gp2 organisms

The listed genera were all detected in sequenced negative controls that were processed alongside human-derived samples in our laboratories (WTSI, ICL and UB) over a period of four years. A variety of DNA extraction and PCR kits were used over this period, although DNA was primarily extracted using the FastDNA SPIN Kit for Soil. Genus names followed by a superscript letter indicate those that have also been independently reported as contaminants previously. [a]also reported by Tanner *et al.* [12]; [b]also reported by Grahn *et al.* [14]; [c]also reported by Barton *et al.* [17]; [d]also reported by Laurence *et al.* [18]; [e]also detected as contaminants of multiple displacement amplification kits (information provided by Paul Scott, Wellcome Trust Sanger Institute). ICL, Imperial College London; UB, University of Birmingham; WTSI, Wellcome Trust Sanger Institute.

Compare abundances in samples and blanks

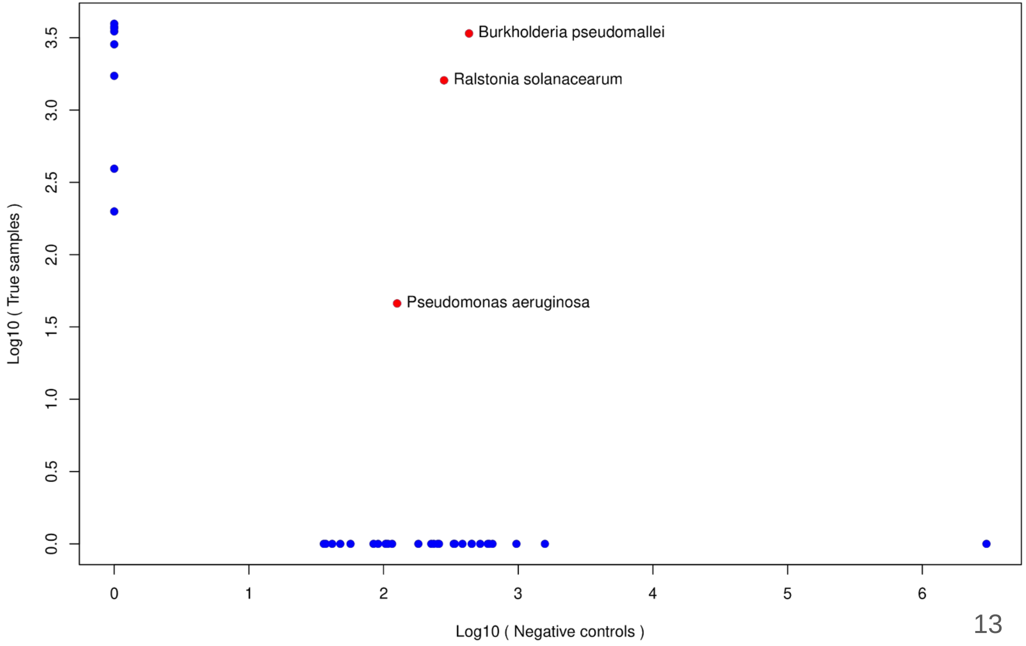
Highlight organisms abundant in both samples and blanks:

```
Rscript blank_decontam.R krakenuniq_abundance_matrix.txt blank_krakenuniq_abundance_matrix.txt
```

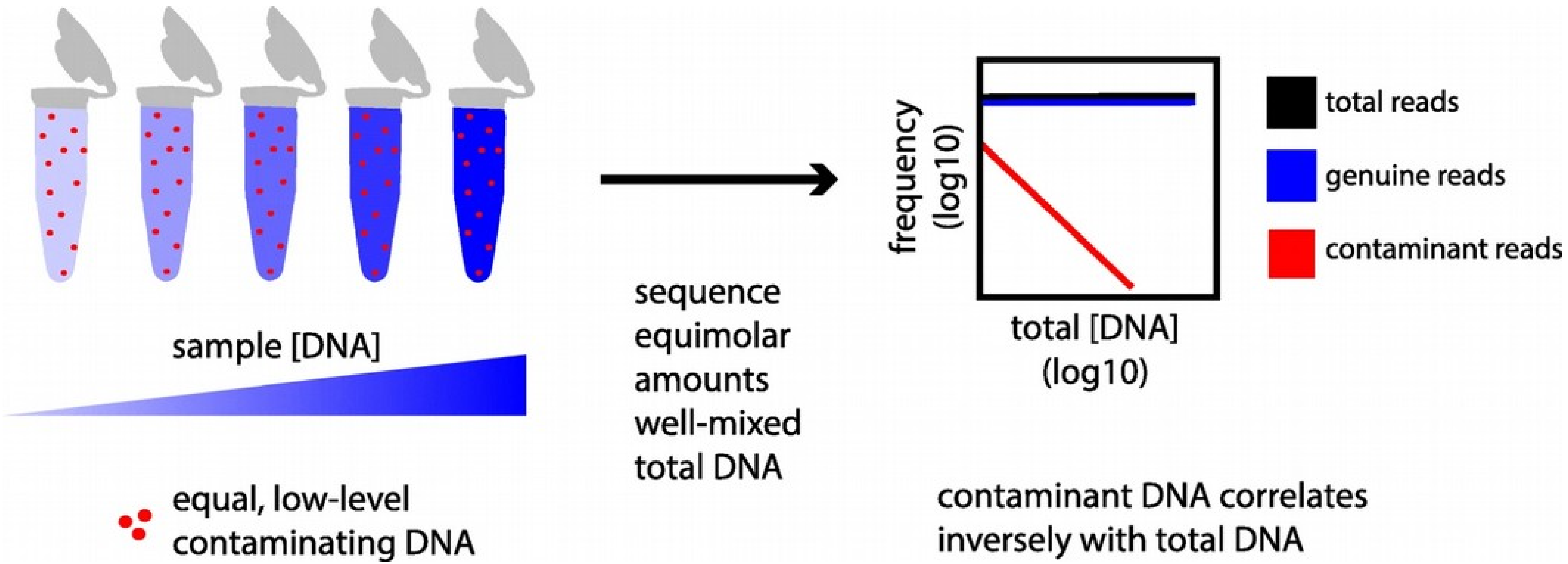
Most organisms are highly abundant either only in sample or only in blanks.

Organisms that are **highly abundant** both in samples and blank negative controls are likely of **exogenous origin**.

Recentrifuge performs similar filtering.

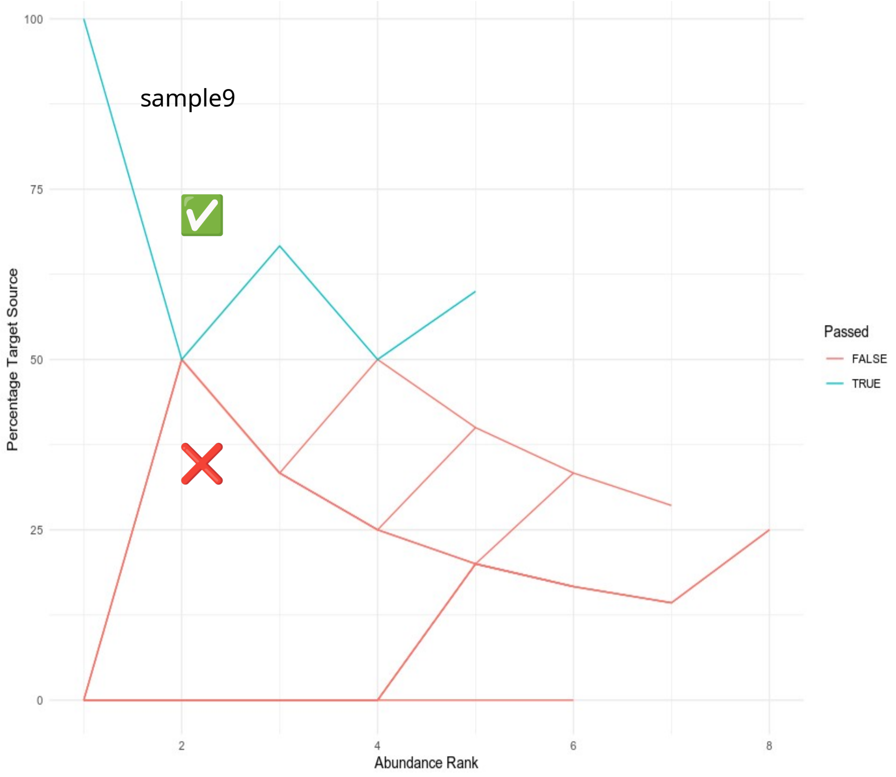


Other approaches: *decontam* R package



Other approaches: *cuperdec* R package

```
library(cuperdec);library(magrittr);library(dplyr)

# Load database (in this case oral database)
data(cuperdec_database_ex)
database <- load_database(cuperdec_database_ex,
target = "oral") %>% print()

# Load abundance matrix and metadata
abundance <- "krakenuniq_abundance_matrix.txt"
taxatable <- load_taxa_table(abundance)
metadata <- as_tibble(data.frame(Sample =
unique(taxatable$Sample), Sample_Source = "Oral"))

# Compute cumulative percent decay curves
curves <- calculate_curve(taxatable, database)
filter_result <- simple_filter(curves,
percent_threshold = 50) %>% print()
plot_cuperdec(curves, metadata = metadata,
filter_result)
```

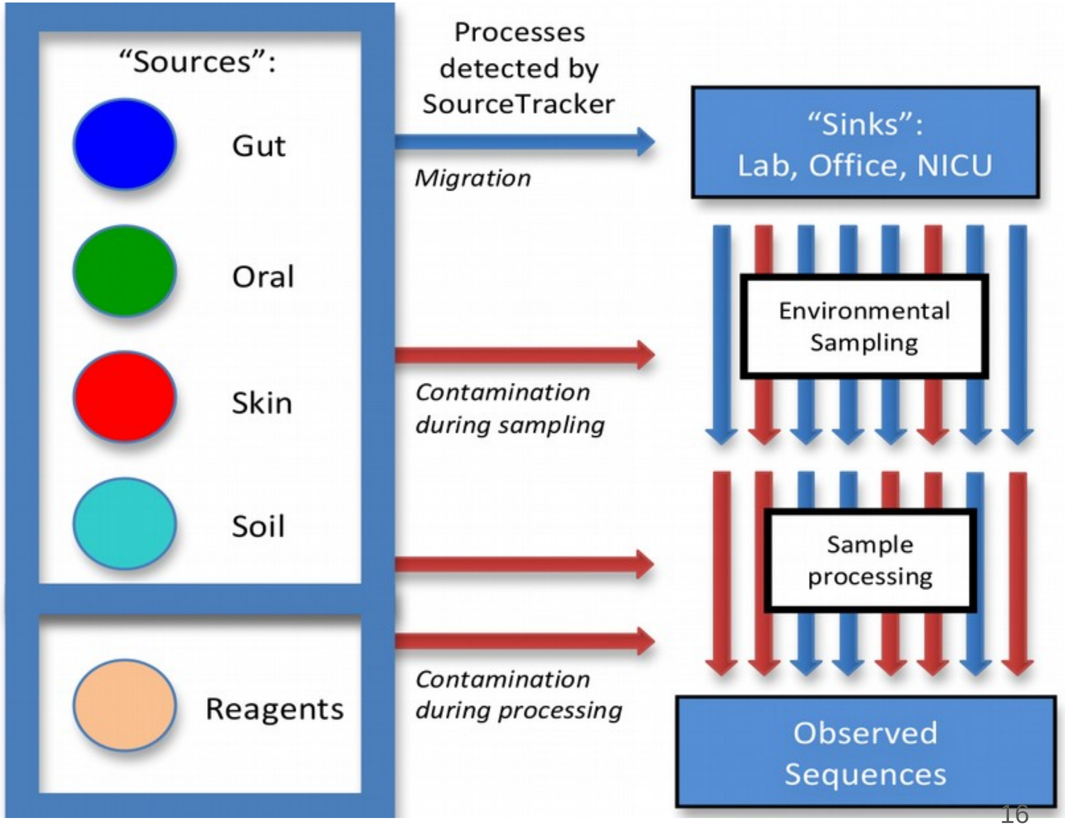


Microbial Source Tracking (MST): SourceTracker

SourceTracker is a Bayesian classifier that infers **environments** that detected microbial organisms originate from.

SourceTracker is trained on **source** samples with known environmental annotation, and makes predictions on **sink** samples.

Extensions are mSourceTracker and FEAST

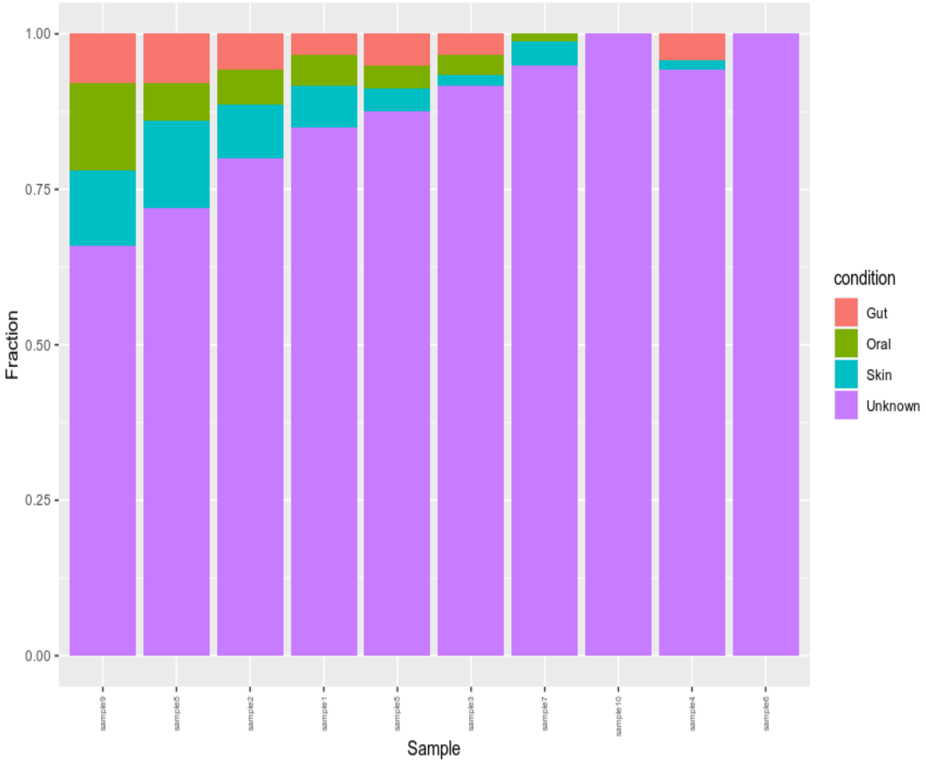


Merge sources and sinks, and train SourceTracker

```
otus_hmp <- read.delim("otus_hmp.txt", header = TRUE, row.names = 1, sep = "\t")
meta_hmp <- read.delim("meta_hmp.txt", header = TRUE, row.names = 1, sep = "\t")

otus_sink<-read.delim("krakenuniq_abundance_matrix.txt", header=TRUE, row.names=1, sep="\t")
otus <- merge(otus_hmp, otus_sink, all = TRUE, by = "row.names")
rownames(otus) <- otus$Row.names; otus$Row.names <- NULL; otus[is.na(otus)] <- 0
meta_sink <- data.frame(ID = colnames(otus_sink), Env = "Unknown", SourceSink = "sink")
rownames(meta_sink) <- meta_sink$ID; meta_sink$ID<-NULL; metadata <- rbind(meta_hmp, meta_sink)

otus <- as.data.frame(t(as.matrix(otus))); otus[otus > 0] <- 1; otus <- otus[rowSums(otus)!=0,]
metadata<-metadata[as.character(metadata$Env)!="Vaginal",]; envs <- metadata$Env
common.sample.ids <- intersect(rownames(metadata), rownames(otus))
otus <- otus[common.sample.ids,]; metadata <- metadata[common.sample.ids,]

# Train SourceTracker on sources (HMP) and run predictions on sinks
source('/sourcetracker/src/SourceTracker.r')
train.ix <- which(metadata$SourceSink=='source'); test.ix <- which(metadata$SourceSink=='sink')
st <- sourcetracker(otus[train.ix,], envs[train.ix])
results <- predict(st, otus[test.ix,], alpha1 = 0.001, alpha2 = 0.001)
```

Plot SourceTracker results

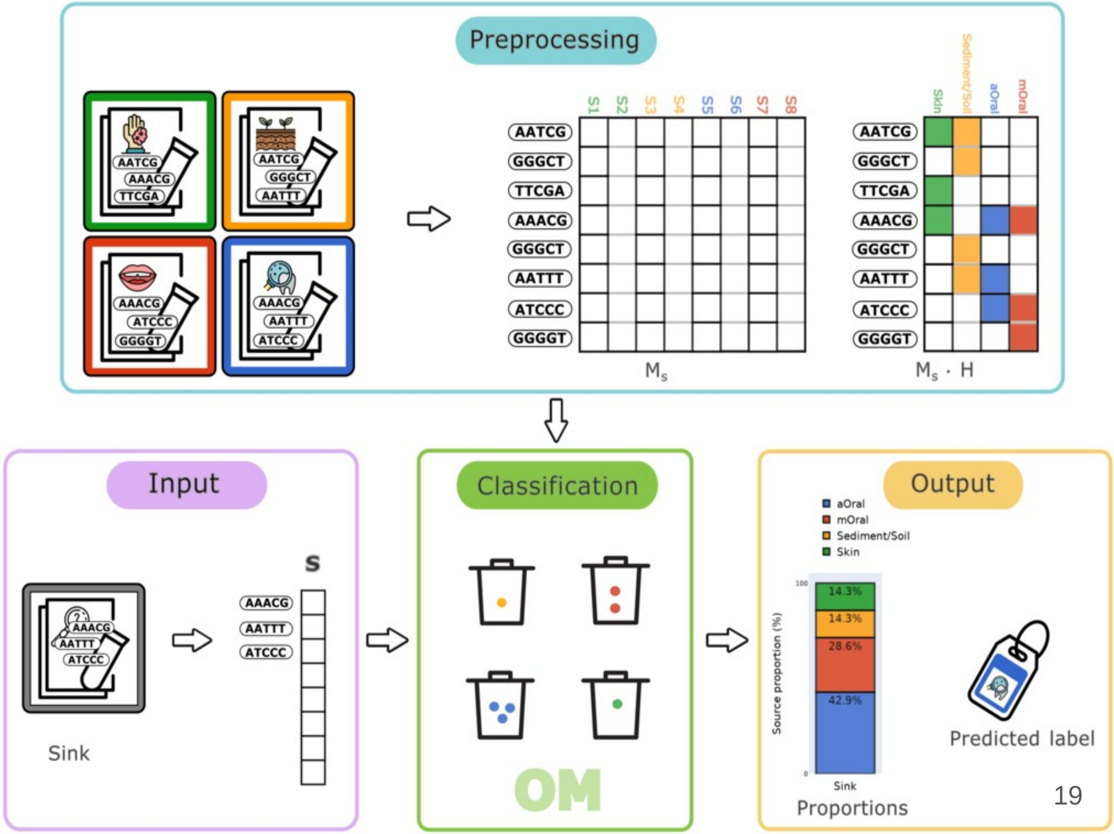
```r
# Sort SourceTracker proportions for plotting
props <- results$proportions
props <- props[order(-props[,"Oral"]),]
results$proportions <- props

# Prepare SourceTracker output for plotting
name<-rep(rownames(results$proportions),each=4)
value <- as.numeric(t(results$proportions))
labels <- c("Gut","Oral","Skin","Unknown")
condition<-rep(labels, length(test.ix))
data <- data.frame(name, condition, value)

# Plot SourceTracker inference as a barplot
library("ggplot2")
ggplot(data, aes(fill=condition, y=value, x =
reorder(name, seq(1:length(name))))) +
geom_bar(position = "fill", stat = "identity") +
theme(axis.text.x = element_text(angle = 90,
size=5,hjust=1,vjust=0.5)) + xlab("Sample") +
ylab("Fraction")
```



Read-level source tracking with decOM

SourceTracker requires a microbial **abundance matrix** computed by e.g. Kraken based on a **taxonomically annotated database**.

In contrast, **decOM** aims at **moving away from database-dependent** methods and using unsupervised approaches exploiting **read-level** sequence composition.



Perform source tracking with decOM

Prepare input *fof*-files that have a key - value format:

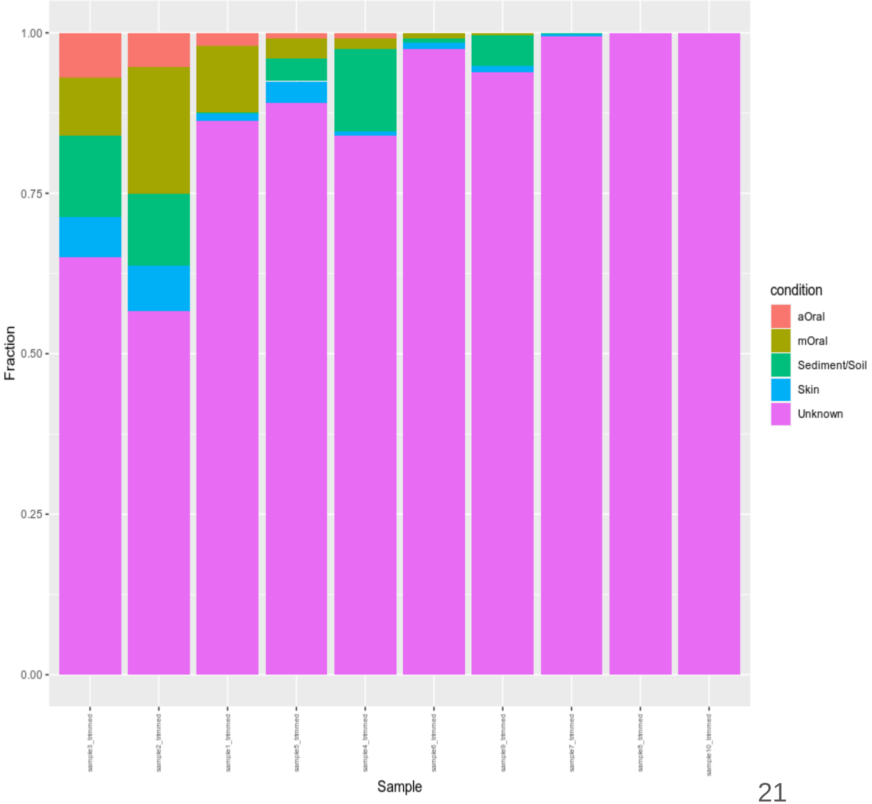
```
cd 03_TRIMMED # folder containing trimmed fastq-files
for i in {1..10}
do
echo "sample${i}_trimmed : sample${i}_trimmed.fastq.gz" > sample${i}_trimmed.fof
echo sample${i}_trimmed >> FASTQ_NAMES_LIST.txt
done
```

Download prebuilt kmer-matrix of sources (aOral,mOral,Sediment/Soil,Skin), run decOM:

```
wget https://zenodo.org/record/6513520/files/decOM_sources.tar.gz  tar -xf decOM_sources.tar.gz

decOM -p_sources decOM_sources/ -p_sinks FASTQ_NAMES_LIST.txt -p_keys CUTADAPT -mem 10GB -t 5
```

Plot decOM source tracking results

```
df<-read.csv("decOM_output.csv",check.names=FALSE)

result<-subset(df,select=c("Sink","Sediment/Soil",
"Skin", "aOral", "mOral", "Unknown"))
rownames(result) <- result$Sink; result$Sink<-NULL
result <- result / rowSums(result)
result <- result[order(-result$aOral), ]

name <- rep(rownames(result), each = 5)
value <- as.numeric(t(result))
condition <- rep(c("Sediment/Soil","Skin","aOral",
"mOral","Unknown"), dim(result)[1])
data <- data.frame(name, condition, value)

library("ggplot2")
ggplot(data, aes(fill = condition, y = value,
x=reorder(name,seq(1:length(name))))) +
  geom_bar(position = "fill", stat = "identity") +
  theme(axis.text.x=element_text(angle=90, size=5,
hjust=1,vjust=0.5))+xlab("Sample")+ylab("Fraction")
```



Summary

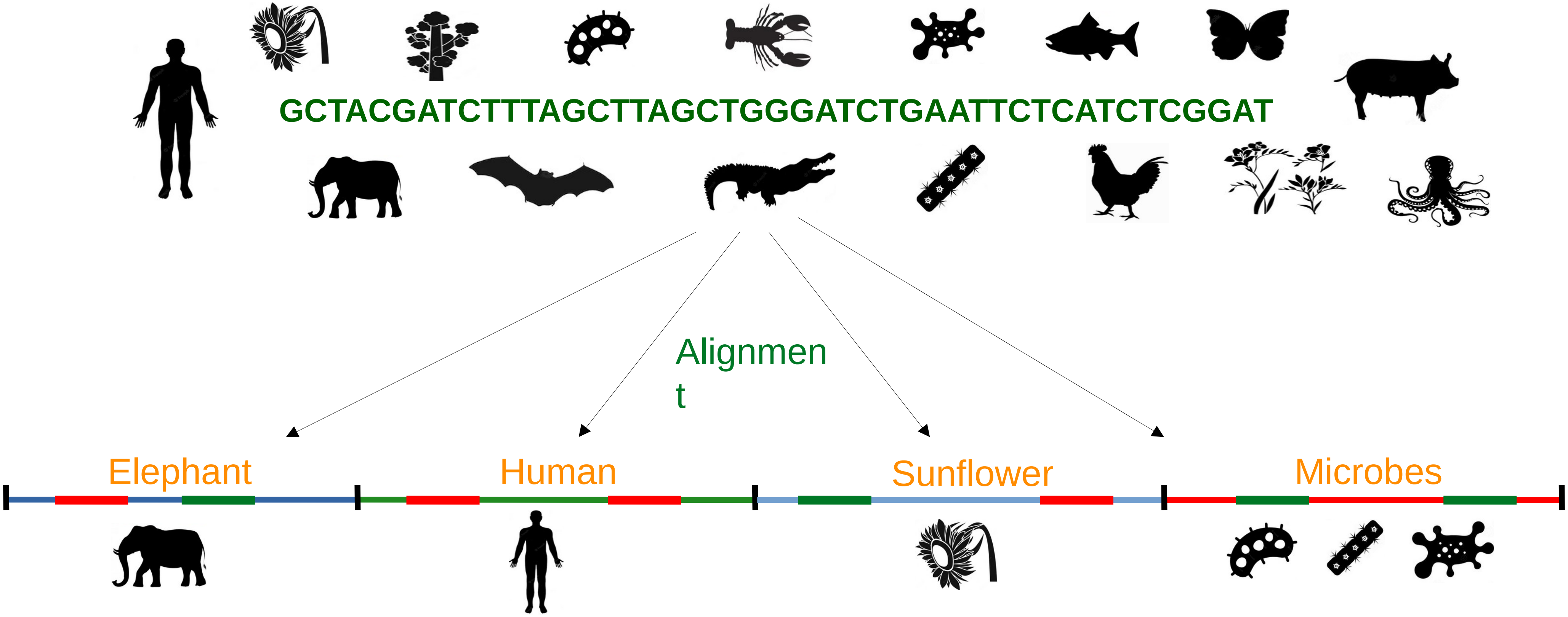
- Negative controls are important for disentangling ancient / endogenous from modern / exogenous contamination

- Microbial source tracking is another layer of evidence that can facilitate interpretation of ancient metagenomic findings

Decontaminating eukaryotic references for ancient <u>environmental</u> metagenomics

Uncertainty with sequence identification

Does it belong to the organism I am thinking about, or is it contamination?



Reference genomes: are they OK or contaminated?

Contamination within reference genomes

NBIS SciLifeLab

Research

Human contamination in bacterial genomes has created thousands of spurious proteins

Florian P. Breitwieser,[1] Mihaela Pertea,[1,2] Aleksey V. Zimin,[1,3] and Steven L. Salzberg[1,2,3,4]

[1]Center for Computational Biology, McKusick-Nathans Institute of Genetic Medicine, Johns Hopkins School of Medicine, Baltimore, Maryland 21205, USA; [2]Department of Computer Science, Whiting School of Engineering, Johns Hopkins University, Baltimore, Maryland 21218, USA; [3]Department of Biomedical Engineering, Johns Hopkins University, Baltimore, Maryland 21218, USA; [4]Department of Biostatistics, Bloomberg School of Public Health, Johns Hopkins University, Baltimore, Maryland 21205, USA

Contaminant sequences that appear in published genomes can cause numerous problems for downstream analyses, particularly for evolutionary studies and metagenomics projects. Our large-scale scan of complete and draft bacterial and archaeal genomes in the NCBI RefSeq database reveals that 2250 genomes are contaminated by human sequence. The contaminant sequences derive primarily from high-copy human repeat regions, which themselves are not adequately represented in the current human reference genome, GRCh38. The absence of the sequences from the human assembly offers a likely explanation for their presence in bacterial assemblies. In some cases, the contaminating contigs have been erroneously annotated as containing protein-coding sequences, which over time have propagated to create spurious protein "families" across multiple prokaryotic and eukaryotic genomes. As a result, 3437 spurious protein entries are currently present in the widely used nr and TrEMBL protein databases. We report here an extensive list of contaminant sequences in bacterial genome assemblies and the proteins associated with them. We found that nearly all contaminants occurred in small contigs in draft genomes, which suggests that filtering out small contigs from draft genome assemblies may mitigate the issue of contamination while still keeping nearly all of the genuine genomic sequences.

OPEN ACCESS Freely available online PLoS one

Abundant Human DNA Contamination Identified in Non-Primate Genome Databases

Mark S. Longo, Michael J. O'Neill, Rachel J. O'Neill*

Department of Molecular and Cell Biology, University of Connecticut, Storrs, Connecticut, United States of America

Abstract

During routine screens of the NCBI databases using human repetitive elements we discovered an unlikely level of nucleotide identity across a broad range of phyla. To ascertain whether databases containing DNA sequences, genome assemblies and trace archive reads were contaminated with human sequences, we performed an in depth search for sequences of human origin in non-human species. Using a primate specific SINE, AluY, we screened 2,749 non-primate public databases from NCBI, Ensembl, JGI, and UCSC and have found 492 to be contaminated with human sequence. These represent species ranging from bacteria (*B. cereus*) to plants (*Z. mays*) to fish (*D. rerio*) with examples found from most phyla. The identification of such extensive contamination of human sequence across databases and sequence types warrants caution among the sequencing community in future sequencing efforts, such as human re-sequencing. We discuss issues this may raise as well as present data that gives insight as to how this may be occurring.

Citation: Longo MS, O'Neill MJ, O'Neill RJ (2011) Abundant Human DNA Contamination Identified in Non-Primate Genome Databases. PLoS ONE 6(2): e16410. doi:10.1371/journal.pone.0016410

Editor: Najib El-Sayed, The University of Maryland, United States of America

Received September 1, 2010; **Accepted** December 23, 2010; **Published** February 16, 2011

Funding: This work was funded by the NSF (www.nsf.gov). The funders had no role in study design, data collection and analysis, decision to publish, or preparation of the manuscript.

Competing Interests: The authors have declared that no competing interests exist.

* E-mail: rachel.oneill@uconn.edu

Current Biology CellPress

Volume 22, Issue 15, 7 August 2012, Pages R593-R594

Correspondence

Origin of land plants revisited in the light of sequence contamination and missing data

Simon Laurin-Lemay, Henner Brinkmann, Hervé Philippe ✉

EMBO reports

Science & Society

Here, there, and everywhere

From PCRs to next-generation sequencing technologies and sequence databases, DNA contaminants creep in from the most unlikely places

Karl Gruber

PNAS CrossMark SEE COMMENTARY

No evidence for extensive horizontal gene transfer in the genome of the tardigrade *Hypsibius dujardini*

Georgios Koutsovoulos[a], Sujai Kumar[a], Dominik R. Laetsch[a,b], Lewis Stevens[a], Jennifer Daub[a], Claire Conlon[a], Habib Maroon[a], Fran Thomas[a], Aziz A. Aboobaker[c], and Mark Blaxter[a,1]

[a]Institute of Evolutionary Biology, University of Edinburgh, Edinburgh EH9 3FL, United Kingdom; [b]The James Hutton Institute, Dundee DD2 5DA, United Kingdom; and [c]Department of Zoology, University of Oxford, Oxford OX1 3PS, United Kingdom

Edited by W. Ford Doolittle, Dalhousie University, Halifax, Canada, and approved March 1, 2016 (received for review January 8, 2016)

Tardigrades are meiofaunal ecdysozoans that are key to understanding the origins of Arthropoda. Many species of Tardigrada can survive extreme conditions through cryptobiosis. In a recent paper [Boothby TC, et al. (2015) *Proc Natl Acad Sci USA* 112(52):15976–15981], the authors concluded that the tardigrade *Hypsibius dujardini* had an unprecedented proportion (17%) of genes originating through functional horizontal gene transfer (fHGT) and speculated that fHGT was likely formative in the evolution of cryptobiosis. We independently sequenced the genome of *H. dujardini*. As expected from whole-organism DNA sampling, our raw data contained reads from nontarget genomes. Filtering using metagenomics approaches generated a draft *H. dujardini* genome assembly of 135 Mb with superior assembly metrics to the previously published assembly. Additional microbial contamination likely remains. We found no support for extensive fHGT. Among 23,021 gene predictions we identified 0.2% strong candidates for fHGT from bacteria and 0.2% strong candidates for fHGT from nonmetazoan eukaryotes. Cross-comparison of assemblies showed that the overwhelming majority of HGT candidates in the Boothby et al. genome derived from contaminants. We conclude that fHGT into *H. dujardini* accounts for at most 1–2% of genes and that the proposal that one-sixth of tardigrade genes originate from functional HGT events is an artifact of undetected contamination.

tardigrade | blobtools | contamination | metagenomics | horizontal gene transfer

cryptobiotic (24), but serves as a useful comparator for good cryptobiotic species (9).

Animal genomes can accrete horizontally transferred DNA, especially from germ line-transmitted symbionts (25), but the majority of transfers are nonfunctional and subsequently evolve neutrally and can be characterized as dead-on-arrival horizontal gene transfer (doaHGT) (25–27). Functional horizontal gene transfer (fHGT) can bring to a recipient genome new biochemical capacities and contrasts with gradualist evolution of endogenous genes to new function. The bdelloid rotifers *Adineta vaga* (28) and *Adineta ricciae* (29) have high levels of fHGT (~8%), and this has been associated with both their survival as phylogenetically ancient asexuals and their ability to undergo cryptobiosis (28–32). Different kinds of evidence are required to support claims of doaHGT compared with fHGT. Both are supported by phylogenetic proof of foreignness, linkage to known host genome-resident genes, in situ proof of presence on nuclear chromosomes (33), Mendelian inheritance (34), and phylogenetic perdurance (presence in all, or many individuals of a species, and presence in related taxa). Functional integration of a foreign gene into an animal genome requires adaptation to the new transcriptional environment including acquisition of spliceosomal introns, acclimatization to host base composition and codon use bias, and evidence of active transcription (e.g., in mRNA sequencing data) (35, 36).

Another source of foreign sequence in genome assemblies is contamination, which is easy to generate and difficult to sepa-

GENETICS

PeerJ

Unexpected cross-species contamination in genome sequencing projects

Samier Merchant[1,2], Derrick E. Wood[1,3] and Steven L. Salzberg[1,3,4]

[1] Center for Computational Biology, McKusick-Nathans Institute of Genetic Medicine, Johns Hopkins University, Baltimore, MD, USA
[2] Department of Computer Science, Brown University, Providence, RI, USA
[3] Department of Computer Science, Johns Hopkins University, Baltimore, MD, USA
[4] Department of Biomedical Engineering, Johns Hopkins University, Baltimore, MD, USA

ABSTRACT

The raw data from a genome sequencing project sometimes contains DNA from contaminating organisms, which may be introduced during sample collection or sequence preparation. In some instances, these contaminants remain in the sequence even after assembly and deposition of the genome into public databases. As a result, searches of these databases may yield erroneous and confusing results. We used efficient microbiome analysis software to scan the draft assembly of domestic cow, *Bos taurus*, and identify 173 small contigs that appeared to derive from microbial contaminants. In the course of verifying these findings, we discovered that one genome, *Neisseria gonorrhoeae* TCDC-NG08107, although putatively a complete genome, contained multiple sequences that actually derived from the cow and sheep genomes. Our findings illustrate the need to carefully validate findings of anomalous DNA that rely on comparisons to either draft or finished genomes.

Steinegger and Salzberg *Genome Biology* (2020) 21:115
https://doi.org/10.1186/s13059-020-02023-1

Genome Biology

METHOD Open Access

Terminating contamination: large-scale search identifies more than 2,000,000 contaminated entries in GenBank

Check for updates

Martin Steinegger[1,2,3*] and Steven L. Salzberg[2,4,5]

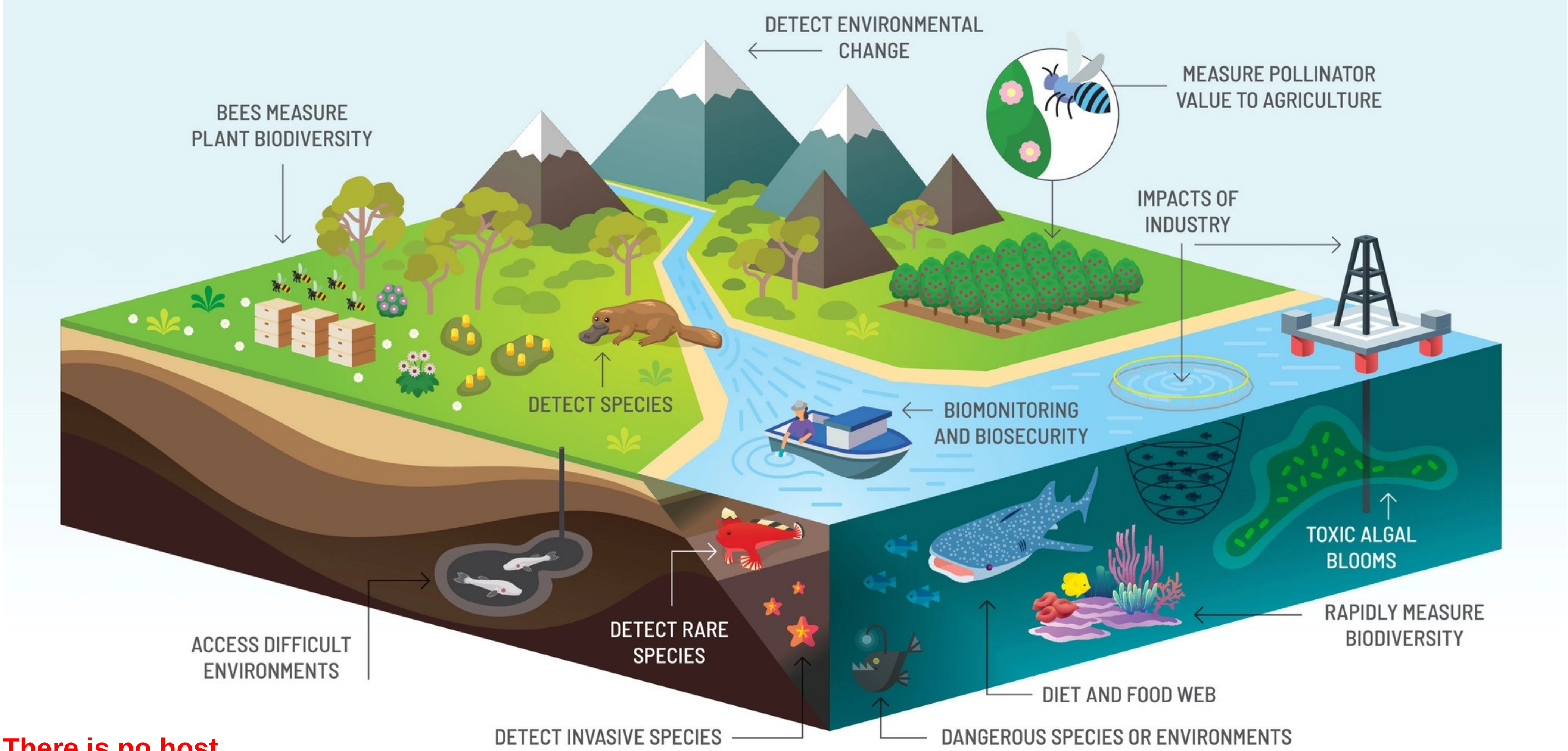
*Correspondence: martin.steinegger@snu.ac.kr
[1]School of Biological Sciences, Seoul National University, Seoul, 08826, South Korea
[2]Center for Computational Biology, Whiting School of Engineering, Johns Hopkins University, 21218 Baltimore, Maryland, USA
Full list of author information is available at the end of the article

Abstract

Genomic analyses are sensitive to contamination in public databases caused by incorrectly labeled reference sequences. Here, we describe Conterminator, an efficient method to detect and remove incorrectly labeled sequences by an exhaustive all-against-all sequence comparison. Our analysis reports contamination of 2,161,746, 114,035, and 14,148 sequences in the RefSeq, GenBank, and NR databases, respectively, spanning the whole range from draft to "complete" model organism genomes. Our method scales linearly with input size and can process 3.3 TB in 12 days on a 32-core computer. Conterminator can help ensure the quality of reference databases. Source code (GPLv3): https://github.com/martin-steinegger/conterminator

Keywords: Genomes, Contamination, Software, RefSeq, GenBank

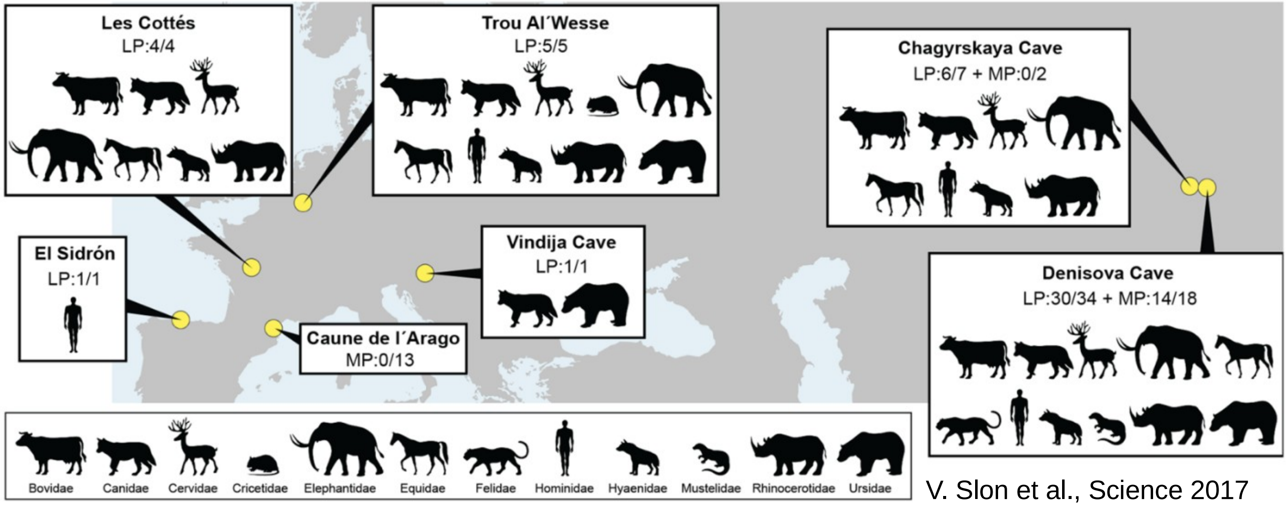
Environmental and sedimentary DNA



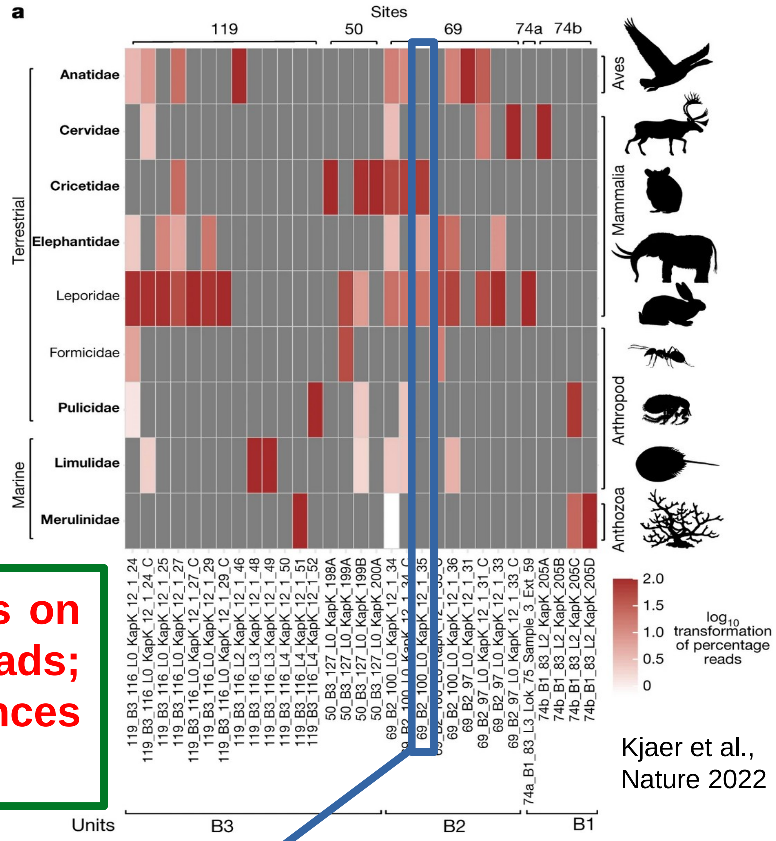
There is no host.
Amount of DNA per organism is limited

Image adapted from https://research.csiro.au/environomics

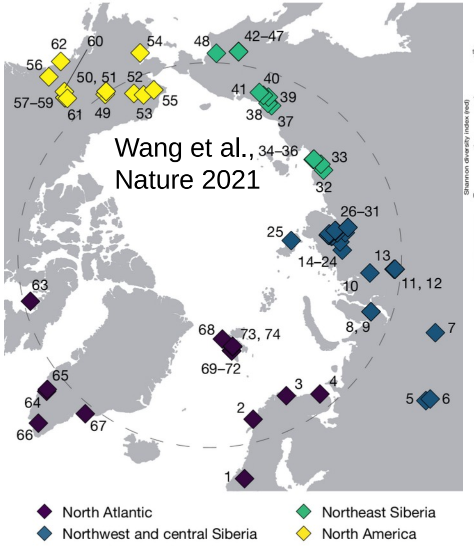
Environmental and sedimentary aDNA

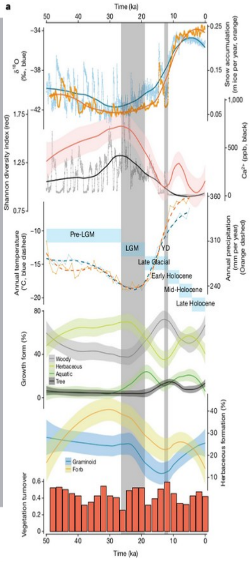


V. Slon et al., Science 2017



Kjaer et al., Nature 2022

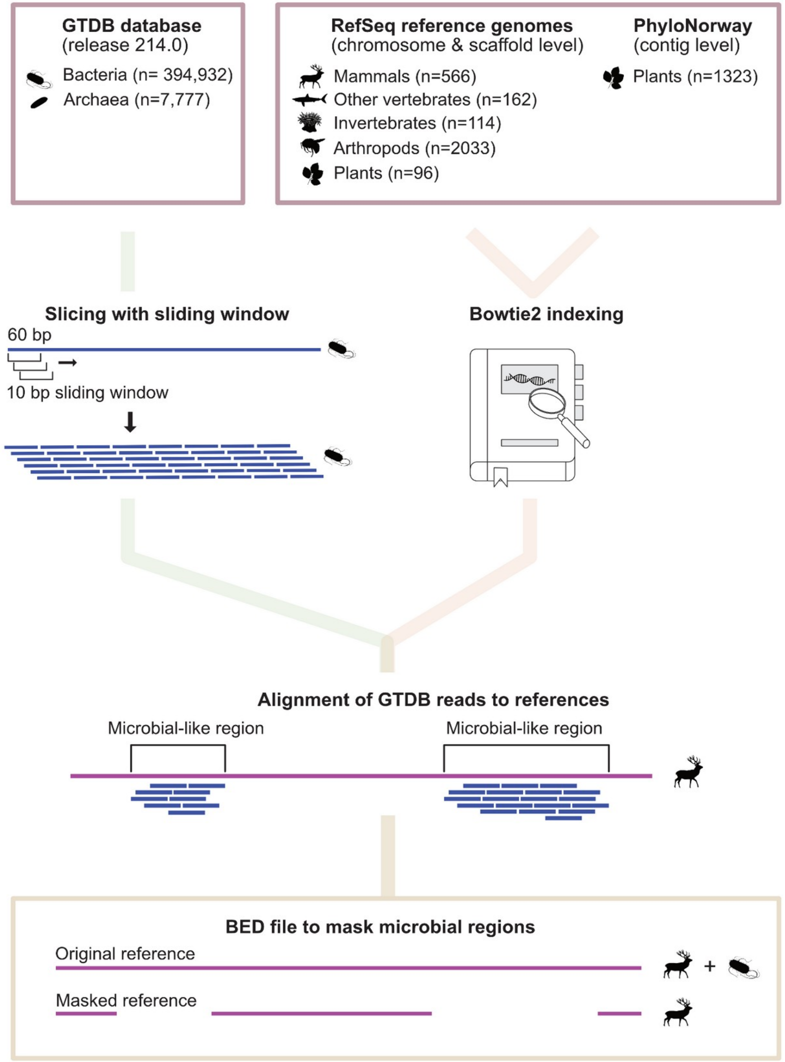


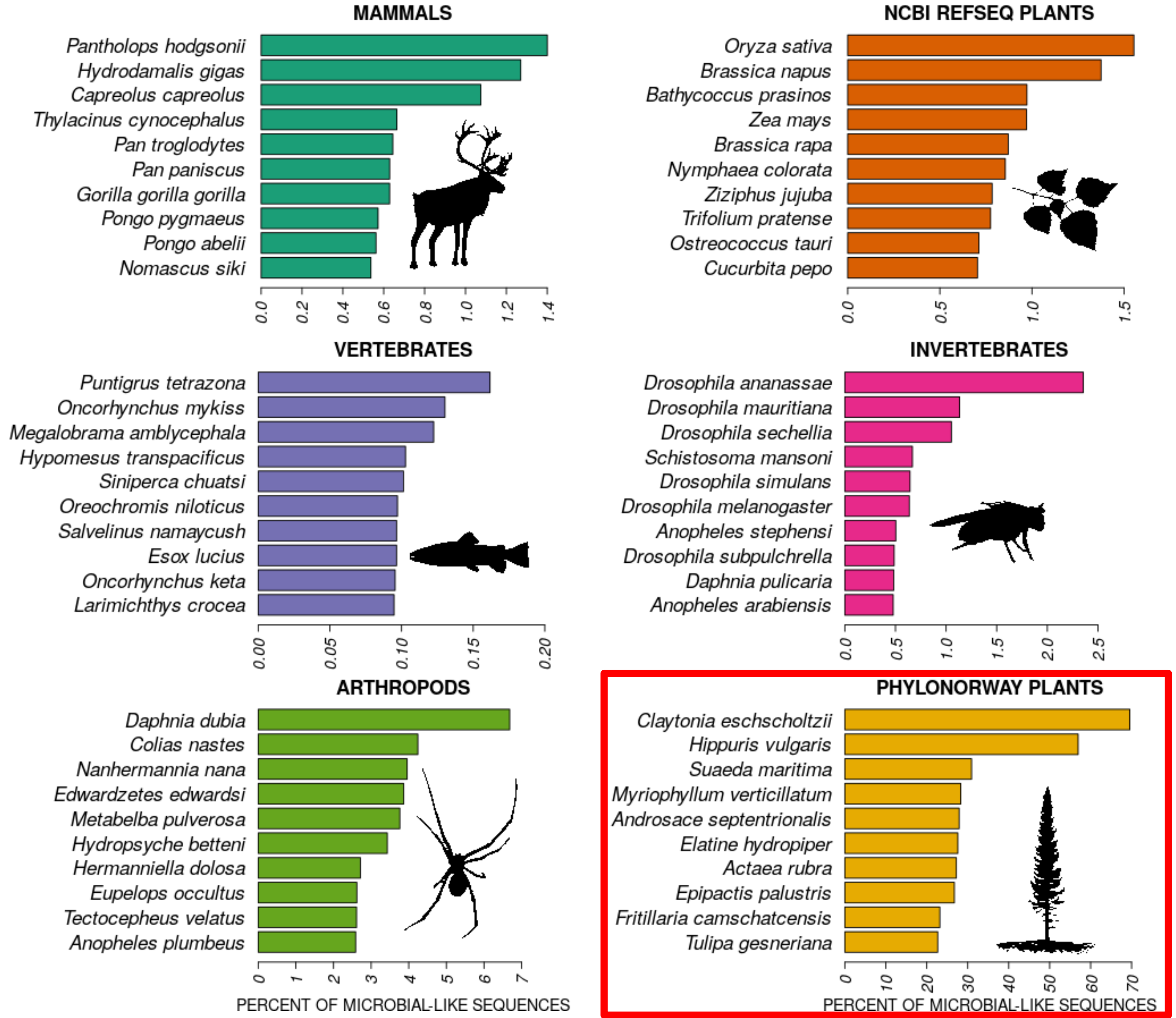


If organism detection relies on a very limited number of reads; hence, high-quality references are particularly important!

A	S	T	U	V	W
Taxa	69_B2_100_L0_KapK_12_1_34	69_B2_100_L0_KapK_12_1_34_C	69_B2_100_L0_KapK_12_1_35	69_B2_100_L0_KapK_12_1_35_C	69_B2_100_L0_KapK_12_1_36
Anatidae	372	3	0	0	70
Cervidae	84		0	0	0
Cricetidae	1331	17	1890	0	0
Elephantidae	85		171	48	118
Formicidae	0		0	19	0
Leporidae	542	5	549	61	313
Limulidae	66		0	0	23
Merulinidae	18		0	0	0
Pulicidae	0		0	0	0

Microbial-like sequences in eukaryotes





Hippuris vulgaris hit in 2 Nature papers

Wang et al., Nature 2021

Article

Late Quaternary dynamics of Arctic biota from ancient environmental genomics

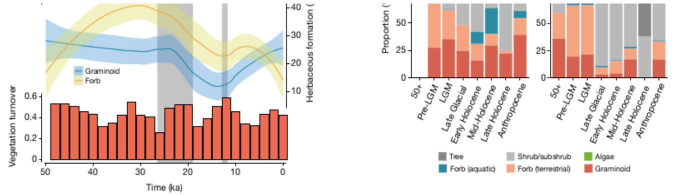


Fig. 2 | Climate and vegetation changes over the past 50 kyr. a, Pan-Arctic climate changes and vegetation variations. LGM (26.5–19 ka) and Younger Dryas (YD) (12.9–11.7 ka) are indicated by grey bars. The six time intervals are indicated by light blue bars (Supplementary Information 2). The error bands denote s.e. From top to bottom (see Methods for detailed calculations): the Greenlandic ice-core $\delta^{18}O$ ratio and snow accumulation rate; the plant Shannon diversity and the Greenlandic ice-core calcium concentration; the average modelled annual temperature and precipitation for all eDNA sampling sites; the proportion of plant growth forms; the proportion of the herbaceous plant growth forms; and the vegetation turnover rates. **b**, The number of observed genera in different regions. **c**, Regional vegetation turnovers. **d**, Regional vegetation morphological compositions. The sample sizes for each region and time interval are provided in Supplementary Information 2. Calculations are supplied in the Methods.

Regional vegetation dynamics

Underlying the generalized pattern of Holarctic vegetation changes are significant geographical differences. Early in postglacial times, the North Atlantic experienced the sharpest rises in taxonomic richness (Fig. 2b), along with the steepest temperature increase (Extended Data Fig. 2b). The increase in postglacial richness was probably driven by species dispersals coupled with habitat diversification[17], that is, gynomorphically dynamic substrates that were exposed by glacial retreat and shaped by meltwater. The resultant vegetation initially had low diversity but was rich in aquatic taxa (Fig. 2b, d). The abundance of aquatic taxa relates in part to the prevalence of samples from lakes in the North Atlantic (Supplementary Information 10), but nonetheless highlights the ability of aquatic plants to disperse rapidly into newly deglaciated terrain containing abundant streams and lake basins[18]. As the postglacial climate continued to warm, the overall proportion of aquatic taxa declined as trees and shrubs (for example, *Betula*, *Salix* and *Vaccinium*) became abundant in this region (Fig. 2d and Extended Data Fig. 3).

Northeast Siberia and North America experienced less radical postglacial changes in vegetation type (Fig. 2c, d). During the Late Glacial, trees and shrubs became more widely distributed, and floristic diversity started to decline—a trend that was especially pronounced in North America (Fig. 2b, d). By about 12 ka, rising sea levels had flooded the Bering Strait, and the vegetation on each side started to diverge (Extended Data Fig. 2a). In northeast Siberia, greater effective moisture

within the Holocene led to the expansion of aquatic plants (such as *Hippuris* and *Menyanthes*). The previously dominant steppe taxa (for example, *Poa* and *Artemisia*) declined, although sedges, of which many species are hygrophilous, continued to be abundant (Extended Data Fig. 3). The vegetation of this region became a mosaic of steppe and tundra elements. In North America, trees such as *Populus* and *Picea* became more widespread during the Early Holocene and previously widespread steppe species declined (Fig. 2d and Extended Data Fig. 3). A broad, southern swath of eastern Beringia became boreal forest.

In contrast to the changes observed in these regions, vegetation in and large (horse and mammoth) mammals, indicating that a wetter environment with a high proportion of hygrophilous plants (that is, moisture-loving plants) was a key factor restricting animal distributions. The distribution of mammoths tends to be positively affected by plant NMDS2, which mainly reflects the proportion of woody plants (particularly shrubs and subshrubs) as opposed to herbaceous plants, whereas the reverse is true for horses (Fig. 3). We also found that horses are more sensitive to vegetation composition compared with other herbivores (Supplementary Information 13.3). These findings support the hypothesis that horses were more restricted to a grassland environ-

Kjaer et al., Nature 2021

nature portfolio

https://doi.org/10.1038/s41586-022-05453-y

Supplementary information

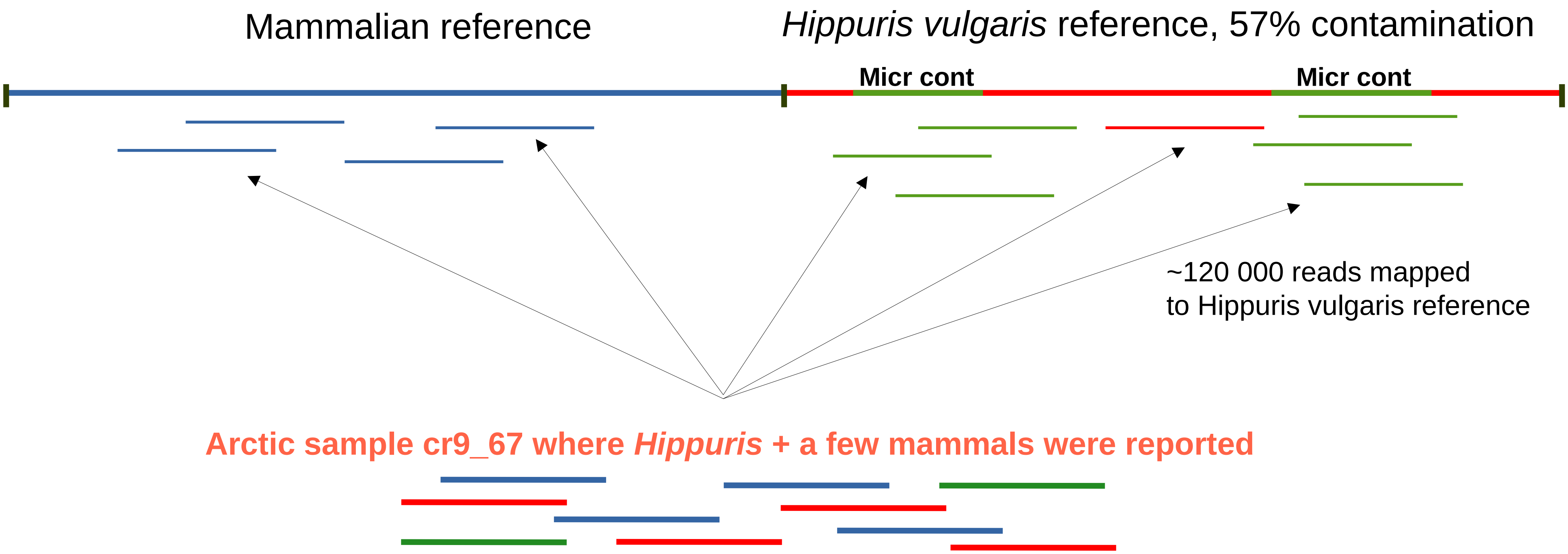
A 2-million-year-old ecosystem in Greenland uncovered by environmental DNA

In the format provided by the authors and unedited

individual samples is consistent with an assemblage from a large low-gradient alluvial catchment, consistent with an estuary with multiple tributaries. All but the two least diverse samples contain both aquatic and terrestrial taxa. The most common terrestrial genera *Andromeda, Salix*, *Vaccinium*, *Carex*, *Dryas* and *Equisetum*, dominate the DNA assemblages in almost all samples and are well represented as macrofossils (Source Data S1 - sheet 2). Most of the aquatic taxa are restricted to standing water (e.g., *Hippuris*, *Stuckenia, Potamogeton, Menyanthes*) or occasionally streams (*Sparganium, Callitriche*). This range of genera combined with the presence of ferns and club mosses is consistent with alluvial deposition from drainage containing a range of microhabitats including well-drained slopes, stream valleys, lakes, coastal plains, and beaches. If the requirements and ecological amplitudes of these genera are generally conserved, the occurrence of *Erica*, *Sphagnum*, *Arctostaphylos* and *Kalmia* would suggest the presence of acidic histosols and snow cover; *Alnus* and perhaps *Populus and Salix* saturated alluvium / riparian habitats; *Artemisia* and *Astragalus* open, primarily mineral soils and *Dryas* cold, open and windswept areas.

A scenario in which eroded sediment from paleosols or permafrost are redeposited after mixing with younger entrained material could produce an assemblage comprising taxa from a succession of communities adapting over millennia to changing conditions. However, we assert that the biotic assemblages of DNA, macrofossils, and pollen from any discrete sample in the Kap København Formation are largely contemporaneous. Redeposited material is unlikely to contribute substantially to the assemblages for several reasons. The macrofossils were generally well preserved,

How much is *Hippuris vulgaris* hit affected?



116 000 reads out of 120 000 mapped reads,
i.e. **97%** of reads, intersect with regions of microbial contamination in Hippuris reference

National Bioinformatics Infrastructure Sweden (NBIS)

Knut och Alice Wallenbergs Stiftelse

Vetenskapsrådet

LUNDS UNIVERSITET